VIKRAM EJJAGIRI

Industrial AI · Machine Learning · Optimization · Digital Production

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PROFILE

Master's student in Applied AI for Digital Production Management (M.Eng.) at TH Deggendorf with a Bachelor's in Mechanical Engineering (Honors in Robotics). I build industrial AI end to end: machine data (OPC UA, MQTT), production optimization, predictive maintenance, visual inspection and LLM assistants. **Available immediately** for a working student position, an internship, a Master's thesis or a full-time role.

EDUCATION

03/2025 – present

M.Eng. Applied AI for Digital Production Management

Technische Hochschule Deggendorf, Germany
Current grade: 2.3 (German scale, 1.0 = best) · 90 ECTS, taught in English
Modules: Machine Learning & Deep Learning in Production and Logistics · Advanced Statistical Methods & Optimization (SPC, design of experiments) · Production Data Management, ERP & MES · Big Data, MLOps & NLP · Cyber-Physical Production Systems & Simulation (AnyLogic) · Production and Logistics Management (Lean, Six Sigma) · Quality & Sustainability (FMEA, process capability) · AI and production-systems case studies

08/2021 – 05/2024

B.Eng. Mechanical Engineering (Honors in Robotics)

Vasavi College of Engineering, Hyderabad, India
CGPA 8.17/10 · Bachelor's thesis: Modified SCARA robot · Honors thesis: Design and analysis of an ornithopter

2018 – 2021

Diploma in Mechanical Engineering

Government Polytechnic, Warangal, India
CGPA 9.52/10

PROJECTS

Kaizen Matrix: my own industrial AI project series (2026), each with code, tests and documentation on GitHub. Prototypes on simulated or public data, not yet validated on a real line

Kaizen Optimize: Production Optimization on Real Data — Bayesian opt. · OR-Tools · SciPy · Streamlit
github.com/vikram-ejjagiri-ai/kaizen-optimize

- Four plant decisions on real public data: cutting data from a model calibrated on C45 lathe tests (Bayesian optimization vs DOE, −40 % machining cost) and job sequencing on a published benchmark (2.6 % from best known).
- Censored Weibull fit on 23,550 SCANIA trucks (best replacement age, −24 % cost vs run to failure) and (s, Q) spare-part policies tested on 12 unseen months; Streamlit dashboard, number-checked LLM advisor, 18 tests.

Kaizen Vision: Comparing Machine Vision Methods for Quality Inspection — OpenCV · PyTorch · scikit-learn · OPC UA
github.com/vikram-ejjagiri-ai/kaizen-vision

- Test bench for choosing a station's vision system: classical features + SVM, CNN features, fine-tuned CNN and PatchCore on real steel-surface and casting images under 44 changes of light, focus, motion, noise and camera position: all 99–100 % when clean, classical 49 % vs fine-tuned CNN trained with variation 94–97 % when changed.
- OPC UA handshake with a simulated PLC (trigger, result, pick position, fail-safe, watchdog, heartbeat; 36 ms), ChArUco webcam calibration (pixel to mm), code readers (QR, Data Matrix, OCR), EN/DE test report, 20 tests.

Production Digitalization Platform: Sensor to Dashboard — MQTT · TimescaleDB · Docker · scikit-learn · SimPy
github.com/vikram-ejjagiri-ai/production-digitalization-platform

- One-click shopfloor platform (Docker Compose): real compressor data (MetroPT-3) and a simulated RFID assembly line (SimPy) over MQTT into TimescaleDB, with data harmonization and RFID de-duplication.
- KPI engine (OEE, FPY, MTBF/MTTR, RFID lead time), health model (Isolation Forest + SPC) with an alarm about 2.5 h before the first reported failure, AI advisor over FMEA/SOPs, Grafana dashboards as code; 20 unit tests.

Kaizen Assist: GenAI Knowledge Assistant for Engineering RAG · Text-to-SQL · DuckDB · Ollama · Streamlit

github.com/vikram-ejjagiri-ai/kaizen-assist

- Hybrid retrieval (BM25 + embeddings, RRF) over specifications, test procedures and PDF test reports with version awareness; cited answers from a local LLM (Ollama) or an API.
- Guarded Text-to-SQL on DuckDB (sqlglot guard, read-only), spec-limit vs measured-value check, prompt-injection guardrails; 96.9 % on 64 development questions, 75 % on a second domain without tuning; 25 unit tests.

KnowFlow AI: Personal AI Assistant for Documents and Spreadsheets FastAPI · Next.js · TypeScript · RAG · OCR

github.com/vikram-ejjagiri-ai/personal-ai-assistant

- Full-stack workspace (FastAPI, Next.js/TypeScript): chat with PDFs, scans (OCR), Word, PowerPoint and Excel files; answers cite page or cell, spreadsheet questions run as validated pandas queries with charts.
- Safe AI edits (preview, confirm, verify, undo; originals unchanged), private accounts per user, version history, LLM fallback chain (Groq, Gemini, Claude, Ollama); 42 tests incl. two-user isolation.

OEE Dashboard & Maintenance Copilot Python · pandas · Streamlit · Ollama · RAG

github.com/vikram-ejjagiri-ai/kaizen-oee-maintenance-copilot

- OEE engine per ISO 22400 with MES data-quality checks (13 weeks: OEE 76.8 %) and a RAG copilot with a local LLM (Ollama) over manuals and work orders; 95.9 % on 49 test questions.

Visual Quality Inspection with Anomaly Detection PyTorch · ONNX Runtime · Streamlit

github.com/vikram-ejjagiri-ai/kaizen-visual-quality-inspection

- PatchCore from scratch in PyTorch on defect-free MVTec AD images (image AUROC 1.000); cost-based threshold caught 46 of 47 unseen defects with no good part rejected; heatmap review app, ONNX export.

Predictive Maintenance: Failure Classification for a CNC Milling Machine scikit-learn · SHAP · Power BI · DAX

github.com/vikram-ejjagiri-ai/kaizen-machine-failure-prediction

- Failure classification on AI4I 2020 with physics-based features (error cost about −⅓): random forest 82 % recall at 93 % precision vs expert rules, SHAP explanations, Power BI what-if.

Industry 4.0 Data Platform OPC UA · MQTT · PostgreSQL · Grafana

github.com/vikram-ejjagiri-ai/kaizen-factory-stack

- Simulated bottling line (7 sensors) as OPC UA server, OPC UA → MQTT bridge, PostgreSQL, live Grafana dashboard.

SKILLS

Programming & data	Python, SQL, pandas, NumPy, Git, Pytest, Playwright, SQLite, DuckDB
Machine learning & vision	scikit-learn (random forest, Isolation Forest), PyTorch, OpenCV, camera calibration (ChArUco), robustness testing, PatchCore anomaly detection, SHAP explainability, ONNX Runtime, feature engineering, cost-based thresholds
Optimization & statistics	Bayesian optimization (Gaussian processes), DOE and response surfaces, scheduling with OR-Tools (CP-SAT) and local search, (s, Q) inventory policies, discrete-event simulation (SimPy), Weibull and survival analysis (censored data, Kaplan-Meier), bootstrap, SPC, SciPy
GenAI & LLMs	RAG, hybrid search (BM25 + embeddings, RRF), Text-to-SQL (sqlglot), local LLMs (Ollama), Groq, Gemini, Claude and OpenAI APIs, guardrails, LLM evaluation, OCR, pypdf
Apps & deployment	FastAPI, Next.js/TypeScript, Streamlit, Plotly, Docker Compose
Industry 4.0 & BI	OPC UA (asyncua), MQTT (Mosquitto), PostgreSQL/TimescaleDB, Grafana, Power BI/DAX, RFID, MES and maintenance data
Manufacturing	OEE (ISO 22400), FPY, MTBF/MTTR, predictive maintenance, visual quality inspection, cutting data and tool life (Taylor), production scheduling, spare-parts planning
Design & automation	SolidWorks, Fusion 360, AutoCAD, MSC Adams, PLC (basic), ROS (basic)
Languages	English (professional) · German (B1, telc certificate 2026) · Telugu (native)

FURTHER INFORMATION

Leadership	President of the Robotics Club, Vasavi College of Engineering (from 12/2023); organised line-follower and maze-solver robot workshops
Award	State rank 43 in the Engineering Common Entrance Test (ECET), 2020
Training	Advanced Robotics – NPTEL (2024) · HVAC Design – KG MECH (2022) · Lathe training – IRA OM Technologies (2021)